

A Bio-Inspired Minimal Model for Non-Stationary K-Armed Bandits

Krubeal Danieli¹ and Mikkel Elle Lepperød²

¹Center for Integrative Neuroplasticity, FYSCELL, University of Oslo, Norway

²Simula Research Laboratory, Oslo, Norway

Abstract

While reinforcement learning algorithms have made significant progress in solving multi-armed bandit problems, they often lack biological plausibility in architecture and dynamics. Here, we propose a bio-inspired neural model based on interacting populations of rate neurons, drawing inspiration from the orbitofrontal cortex and anterior cingulate cortex. Our model reports robust performance across various stochastic bandit problems, matching the effectiveness of standard algorithms such as Thompson Sampling and UCB. Notably, the model exhibits adaptive behavior: employing greedy strategies in low-uncertainty situations while increasing exploratory behavior as uncertainty rises. Through evolutionary optimization, the model’s hyperparameters converged to values that align with known synaptic mechanisms, particularly in terms of synapse-dependent neural activity and learning rate adaptation. These findings suggest that biologically-inspired computational architectures can achieve competitive performance while providing insights into neural mechanisms of decision-making under uncertainty.

1 Introduction

The ability to make decisions for long-term reward maximization is a fundamental aspect of cognition. The brain has evolved specialized and interconnected regions to implement this behaviour under the constraints of biology.

Well-studied ecological settings of decision-making are foraging tasks, such as food search. In these problems, the agent is usually asked to choose between different options to maximize an expected reward. In nature, animals have been shown to exhibit different strategies depending on context. *Matching behaviour*

is a well-known phenomenon in which the animal’s decision patterns are proportional to the reward probability of the available options. Such behaviour is thought to result from the trade-off between exploration and exploitation [1, 2]. In fact, this is a well known phenomenon in the reinforcement learning literature, in which an agent is faced with the dilemma of exploring new alternatives, potentially more rewarding, or exploiting known options, despite being possibly sup-optimal.

A popular formalization of these type of tasks is the *multi-armed bandit* problem (MAB) [3]. This setting is usually described in terms of a slot machine endowed with K distinct arms, also called levers. During a round, the agent selects one of the arms and collects a reward R according to an unknown reward probability specific to the chosen arm. The goal is simply to maximize the total reward after a given number of steps, which is achieved by effectively updating a selection policy after each round. This problem has been extensively studied in the context of reinforcement learning, and it is considered a fundamental building block for more complex tasks [1].

The multi-armed bandit problem comes in several variants, with the simplest featuring a stationary reward distribution. An important performance measure in these tasks is *regret*, usually defined as the distance between the selected choice and the theoretically optimal one. Researchers have proposed numerous algorithms to address this problem, each with distinct theoretical guarantees.

Thompson sampling (TS) is a widely adopted approach rooted in Bayesian optimization. It maintains a posterior distribution over action reward probabilities and selects actions by sampling from these distributions. Thompson sampling has demonstrated near-optimal regret bounds in stochastic settings [4, 5].

In contrast, the Upper Confidence Bound (UCB) algorithm uses an optimistic principle for exploration. It maintains an estimate of the reward for each option by a confidence interval. Action selection relies on the upper bound of this interval, encouraging exploration of less-visited options by assigning them higher uncertainty. UCB has been shown to achieve logarithmic regret in stochastic bandits [6].

Another effective baseline is the ϵ -Greedy strategy. At each decision step, a random action is selected with probability ϵ , and the best-known action with probability $1-\epsilon$ (exploitation). Although not as theoretically optimal as Thompson sampling or UCB, ϵ -Greedy is simple to implement and often effective in practice. Extensions such as VDBE adapt ϵ dynamically based on the value function’s variance, providing better control over exploration [7, 8, 9, 10].

However, these traditional algorithms, despite their effectiveness, lack biological plausibility – they neither resemble neural circuits nor follow synaptic plasticity dynamics. For instance, they do not rely on a network-like architecture with interconnected units, as seen in the brain. Additionally, their action selection process is typically instantaneous, whereas decision-making in the brain occurs over time, often requiring the activity of a neural circuit to evolve and stabilize before converging on a final selection.

Their learning mechanisms also differ fundamentally. They typically involve

explicit updates to statistical parameters (e.g., reward estimates or exploration rates) based on observed outcomes. In contrast, biological learning relies on local plasticity rules, where synaptic changes depend on the activity of connected neurons, modulating how input is integrated and how output signals are generated.

While not the primary driver, these limitations align with a growing interest in machine learning towards bio-inspired algorithms, such as neural networks and predictive coding [11, 12], offering several advantages.

Indeed, these methods can achieve state-of-the-art performance in various domains, including the challenging *machine-challenging tasks* (MCTs), set of problems that are hard for machines but relatively easy for humans [13, 14, 11]. Further, bio-inspired models enhance algorithmic interpretability by clarifying the functional relationships between internal components. When applied to tasks with existing experimental data, these models can generate new insights into brain and suggest novel research directions [15]. While other approaches such as Bayesian learning can demonstrate optimal performance and can match well human data [16], they are more difficult to relate to neuronal dynamics.

In this work, we aim to enhance the biological plausibility of models used in multi-armed bandit tasks by introducing a novel, minimal decision-making architecture. This model comprises two interacting rate-based neuronal populations connected by plastic synapses, and uses a biologically inspired plasticity rule.

The model’s plasticity mechanism is non-Hebbian and depends on the magnitude of inter-population synaptic weights. This formulation aligns with synapse-type specific plasticity (STSP), a biologically supported mechanism linking learning dynamics to synaptic resource availability, current state, and morphological properties [17, 18, 19, 20]. Learning rates are also adaptive, in line with observations in human experiments [16].

Similar forms of plasticity have been employed in prior work on spiking neural networks and models of synaptic metaplasticity [21, 22]. Despite its simplicity, our model performs comparably to standard algorithms such as Thompson Sampling, ϵ -Greedy, and Upper Confidence Bound, while offering a more neurobiologically grounded account of decision-making.

Other studies have also proposed solutions to bridge reinforcement learning and neural mechanisms. For instance, [23] proposed belief-state temporal difference algorithms [24] that abstract dopaminergic signaling and medial prefrontal projections. These models highlight the role of hidden-state inference during probabilistic tasks, though they assume fixed reward distributions and do not incorporate explicit synaptic plasticity.

In [25], metaplasticity mechanisms were explored in relation to the probability estimation of binary sequences, uncovering informative patterns of functional synaptic states. However, the environment varied along a single stimulus dimension. Similarly, [26] applied a metaplasticity model to a probabilistic reversal learning task—effectively a stationary two-armed bandit—revealing the

emergence of option-specific learning dynamics.

A notable exception is [22], which addressed a non-stationary two-arm bandit using a synaptic cascade model [27] equipped with a surprise detection mechanism to track changes in reward probability.

In contrast to previous work, our study addresses more challenging, high-dimensional non-stationary reward environments—featuring up to one thousand arms with independently drifting reward probabilities. We specifically focus on stochastic bandit problems with *concept drift*, where reward distributions evolve over time, either gradually or through abrupt changes, thus requiring flexible and adaptive decision-making strategies [28, 29, 30].

Despite its simplicity, the proposed model performs competitively with standard algorithms such as Thompson Sampling, ϵ -Greedy, and Upper Confidence Bound, while offering a more biologically grounded perspective on decision-making mechanisms.

Overall, our work aims to bridge adaptive decision-making under uncertainty with principles from computational neuroscience. By proposing a biologically plausible mechanism for choice behavior in dynamic environments, we contribute a framework that may inform both the development of adaptive artificial systems and the interpretation of neural processes underlying flexible behavior.

The remainder of this paper first describes our model design and learning, then presents experimental results and comparative analyses with established algorithms, lastly discusses the findings’ broader implications and potential future directions.

2 Methods

The following section is organized as follows. First, we introduce a formalization of general problem setting, together with the variants considered in this work. Then, we outline the architecture of our model and how it can be mapped to neurobiology. Finally, we describe the learning procedure, and showcase its dynamics in a simple example.

2.1 Binomial MAB problem

The standard formulation of the task is structured as a set of K arms (or levers) $\mathcal{A}_K = \{a_1 \dots a_K\}$, with an associated reward distribution $\mathbf{p} = \{p_1, \dots p_K\}$. At each iteration, the agent pulls an arm and collects a possible reward drawn as a Bernoulli variable $R \sim \mathcal{B}(\{0, 1\}, p_k)$. The agent’s objective is maximizing the total reward $\sum_t^T R_t$, after a certain number of rounds T , also called horizon. Importantly, the agent is unaware of the true reward probabilities, and thus has to make its decisions following a certain policy, denoted as π . In the reinforcement learning literature, the policy is often defined as a distribution over actions, here the arms $\mathcal{A}_K$, given the current state at time t . In the bandit

problem, the state can be taken to correspond to the history h_t of past actions and rewards in the period $(0 \dots t]$, and the policy as a function that return a selected arm $\pi(h_t) = a_t$ [31].

Given the inherent stochasticity of the feedbacks from the environment, the policy is affected by the so-called exploration-exploitation trade-off, which here is phrased as the contrast between the option of the arm with the estimated highest expected reward versus the option to explore other arms, so to gather more information. A common approach is the ϵ -greedy policy, where the choice to explore is selected with a probability ϵ . Moreover, it is often preferable to have a more explorative behaviour early during the training, with the intent to have a good sample size for the empirical reward distribution, which can be later exploited for maximizing reward.

Another important concept in multi-armed bandit problems is *regret*. Intuitively, it quantifies the loss of reward due to following a certain policy, and it is determined by the difference between the collected reward and the theoretical optimal, obtained by choosing the best arm at each round. Formally, given defined a function $r(\pi)$ which returns the expected reward while following policy π , the regret ρ over an horizon T can be formulated as:

$$\rho = \frac{1}{T} \sum_t^T p_t^* - r(\pi(h_t)) \quad (1)$$

where p_t^* is the expected reward of the optimal arm at time t , which correspond to its probability since it is a Bernoulli distribution. The goal of the agent is to minimize the regret, and thus maximize the total reward.

2.2 Model description

The model is constructed as a rate network composed of two neuronal populations, U and V . The first, U , represents the memory traces of the K available options (*i.e.*, the bandits), while the second, V , encodes their values under the current policy.

In our model, the first layer represents the available options, while learned connections to the second layer encode their values based on recent reward history. A key simplification is the lumping of option representations into single neurons. While this choice abstracts the more distributed encoding found in actual brain networks, it allows for a more tractable model design [32].

More formally, the model is defined by a set of coupled ordinary differential equations (ODEs). The first equation describes the evolution of the neural activity $\mathbf{u}$ in population U , while the second governs the activity $\mathbf{v}$ in population V , each evolving with its respective time constant τ .

$$\begin{aligned} \tau_u \dot{\mathbf{u}} &= -\mathbf{u} + \mathbf{W}^{VU} \phi_v(\mathbf{v}) + \mathbf{I}_{\text{ext}} \\ \tau_v \dot{\mathbf{v}} &= -\mathbf{v} + \widetilde{\mathbf{W}}^{UV} \phi_u(\mathbf{u}) \end{aligned} \quad (2)$$

The external input $\mathbf{I}_{\text{ext}}$ is a constant input that is used to set the initial conditions of the neural activity $\mathbf{u}$. The activation functions ϕ_v, ϕ_u are applied to population v and u respectively, and represent two distinct neural response functions tailored to each population vector. They have been chosen to be a step-function with threshold θ_v, θ_u applied to a generalized sigmoid with gain g_v, g_u and offset s_v, s_u .

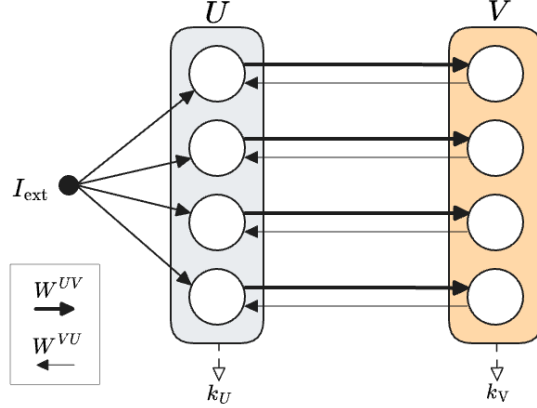


Figure 1: MODEL ARCHITECTURE - The model is composed of a layer U (grey), receiving a feedforward input I_{ext} , a layer V (orange), and connections $\mathbf{W}^{UV}$ and $\mathbf{W}^{VU}$. Additionally, two indexes k_U, k_V are extracted from the layers and corresponds to the selection made by the two populations as $k_U = \text{argmax}_k\{\mathbf{u}\}$, $k_V = \text{argmax}_k\{\mathbf{v}\}$.

Importantly, the two layers are not fully connected and the matrices are diagonal. More in detail, the weight matrix $\mathbf{W}^{VU}$ is simply made of 1s, while $\widetilde{\mathbf{W}}^{UV}$ is a function of the actual weights $\Phi_v(\mathbf{W}^{UV})$ and it represents the contribution of the active options $\mathbf{u}$ to the value representation $\mathbf{v}$, it is thus referred to as *option value function*. The matrix $\mathbf{W}^{UV}$ is initialized to all zeroes. The function Φ_v is defined as weighted sum of a generalized sigmoid and a Gaussian, whose shape is characterized by a bell curve smoothly settling to a constant value. For details see the appendix 5.

The motivation behind our choice of Φ_v is to be agnostic about its final form, and allow competition or integration of two distinct traits of the function shape. In particular, one corresponds to a smooth transition to a plateau value with a certain steepness (or gain), which can represent a saturation once a threshold is crossed, such features has been reported for both biological and artificial neurons [33, 34]. The other is a bell-shaped curve with a defined center and width, which can allow for placing emphasis on values only within a given window and modulate information transfer [35].

The model hyperparameters were optimized for maximizing the average total reward over multiple runs. In particular, given the non-differentiability of the model with respect to the fitness function we employed an evolutionary

algorithm, more specifically CMA-ES.

2.2.1 Option selection

The decision-making process within a single round is structured in two distinct phases. Initially, the model receives a constant external input targeting all neurons in the memory population U equally. During this phase, $\mathbf{I}_{\text{ext}}$ works as an equilibrium value while the reciprocal interactions with population V push $\mathbf{u}$ to different values, depending on the current policy encoded in $\widetilde{\mathbf{W}}^{UV}$. Importantly, the weights $\mathbf{W}^{UV}$ are initialized to zero, and thus the input from U to V is uniform. This approach ensures the absence of biases towards any arm by having all weights equal, and corresponds to a completely untrained network. After a fixed amount of time $\sim 2\text{s}$, the second phase begins. Here, the external input is removed and the model is left to evolve autonomously, and since there are no recurrent connections in neither population the dynamics are entirely driven by their coupling. A selection k is sampled after another fixed amount of time $\sim 5\text{s}$, and it is defined according to the following rule:

$$k = \begin{cases} \operatorname{argmax}_k\{\mathbf{v}\} & \text{if } \operatorname{argmax}_k\{\mathbf{v}\} = \operatorname{argmax}_k\{\mathbf{u}\} \\ \operatorname{random}(K) & \text{otherwise} \end{cases}$$

The selection rule is simple: if the value representation $\mathbf{v}$ is in agreement with the memory trace $\mathbf{u}$, then the option with the highest value is selected. Otherwise, a random option is chosen. This rule is a way to express the exploration-exploitation trade-off, and it is dependent on the current value of the weights $\widetilde{\mathbf{W}}^{UV}$.

Below in subsection 2.2.1, it is reported the pseudo-code for the algorithm behind the selection process, which is applied during each round t .

Lastly, the structure of the option selection process resembles the prefrontal circuitry, as choices emerge from network state sampling following a period of autonomous neural activity. The stability of these neural activations depends on the strength and reliability of the highest-valued option [36, 37].

According to the values of the policy's parameters, the behaviour of the model displays periods of exploration followed by a steady exploitation, which can be reverted in case of a change in the environment's reward distribution.

2.3 Learning

Given a selected option k , the environment (set of bandits) samples and returns a reward $R \in \{0, 1\}$ with probability p_k . Then, the weights $\mathbf{W}^{UV}$ for the neuron corresponding to the option k are updated according to the following plasticity rule:

$$\Delta \mathbf{W}_k^{UV} = \tilde{\eta}_k \left(R \cdot w^+ - \mathbf{W}_k^{UV} \right) \quad (3)$$

Algorithm 1: Two-phases option selection process

Input: External input $\mathbf{I}_{\text{ext}}$, population $\mathbf{u}$, population $\mathbf{v}$, weights $\widetilde{\mathbf{W}}^{UV}$
Output: Selected action k
Phase 1: *external input* ; // Duration: $\sim 2s$
Define constant $\mathbf{I}_{\text{ext}}$;
Update populations $\mathbf{u}, \mathbf{v}$ according to 2.2;
Phase 2: *autonomous evolution* ; // Duration: $\sim 2s$
Remove external input $\mathbf{I}_{\text{ext}}$;
Let system evolve through population coupling according to 2.2;
Selection process:
 $k_u \leftarrow \text{argmax}_k \{\mathbf{u}\};$
 $k_v \leftarrow \text{argmax}_k \{\mathbf{v}\};$
if $k_u = k_v$ **then**
 $k \leftarrow k_v$; // Exploitation
else
 $k \leftarrow \text{random}(K)$; // Exploration
end
return k

where w^+ is a constant maximum synaptic weight, while $\tilde{\eta}_k$ is the learning rate for the option k determined by a function Φ_η of the current weights $\mathbf{W}_k^{UV}$, referred to as *learning rate function*.

The shape of Φ_η is again a Gaussian-sigmoid but with different parameters, giving evolution the opportunity to combine the two characteristic traits of plateau and bell-shaped tuning. In particular, these features can be combined so to define mechanisms of synapse-type specific plasticity as a function of the current synaptic strenght [17], as well the application of other useful homeostatic constraints with computational advantages, such as synaptic scaling and proportional updates [38, 39, 40].

3 Experiments

The model has been tested in a series of benchmark environments, each with a different number of arms and reward distributions. The performance has been compared with the following algorithms: Random Baseline, Upper-Confidence Bound (UCB), Thompson Sampling, and Epsilon-Greedy.

3.1 Game variants

Our goal is to investigate the performance of the agent in a non-stationary environment, meaning that its underlying distribution changes over time ¹ We

¹Since the arm probabilities are not normalized to 1, it is technically improper to call them *probability distributions*; we will therefore refer to either *probability* or *distribution* separately

choose this setting as it resembles an scenario in which an animal has to forage in an environment with food (reward) distributed over a set of fixed locations, but whose occurrence probability can change over time. A *round* -or horizon- is defined as an action-reward event; instead, a *trial* is a block of rounds. For testing, four slightly different MAB variants were used, obtained by introducing different types of non-stationarity: piecewise constant, uniformly changing, sinusoidally changing, and sinusoidally changing with piecewise constant arms. The reason for these choices is to test the model performance under different speed and uniformity of the distribution changes. Figure 2 visually illustrates their specificities.

Piecewise stationary distribution [MAB-P]

Within a trial the reward distribution is stationary and it is drawn from a normal $\mathbf{p} = \mathcal{N}(0.5, 0.2)^K$, clipped in $(0, 1)$. At the end of each trial i it is drawn a new distribution $\mathbf{p}_i \rightarrow \mathbf{p}_{i+1}$ [31].

Piecewise stationary distribution with drift [MAB-D]

At the very beginning, the reward distribution $\mathbf{p}$ is sampled from a normal $\mathbf{p} = \mathcal{N}(0.5, 0.2)^K$. Then, it changes gradually over the rounds, tracked as time t , such that its values tend towards a target distribution $\mathbf{q}_i$ as $\tau_p \dot{\mathbf{p}}_t = \mathbf{q}_i - \mathbf{p}_t$. Here, $\dot{\mathbf{p}}$ is the time derivative of the distribution and τ_p is its time constant. Once the distance is below a threshold δ as $|\mathbf{q}_i - \mathbf{p}_t| < \delta$, the target distribution is changed to a new one $\mathbf{q}_i \rightarrow \mathbf{q}_{i+1}$. In this variant, there are no proper trials but the target distribution keep changing until a maximum number of rounds is reached.

Sinusoidal distribution shift [MAB-sin]

The reward distribution changes over rounds, with the probability of each arm following a sine wave with a specific frequency f_k , phase λ_k and amplitude 1. At any given time t , the distribution is $\mathbf{p}_t = \{\sin(2\pi f_k t + \lambda_k)\}$ for $k = 1 \dots K$.

Partial sinusoidal distribution shift [MAB-sinP]

Identical to the sinusoidal distribution shift, but a half of the arms change sinusoidally while the other half is always kept at a constant value. The distribution is not normalized.

3.2 Evolution search

The optimization of the hyper-parameters was performed using the Covariance Matrix Adaptation evolutionary strategy algorithm (CMA-ES) [41]. The search was run with a population of 256 individuals for 80 generations. Each individual was endowed with a genome, corresponding to a vector of 22 parameters of the model. The fitness function of the evolution was defined as the average reward obtained by an individual over 3 different non-stationary bandit environments,

at any given time for avoiding confusion.

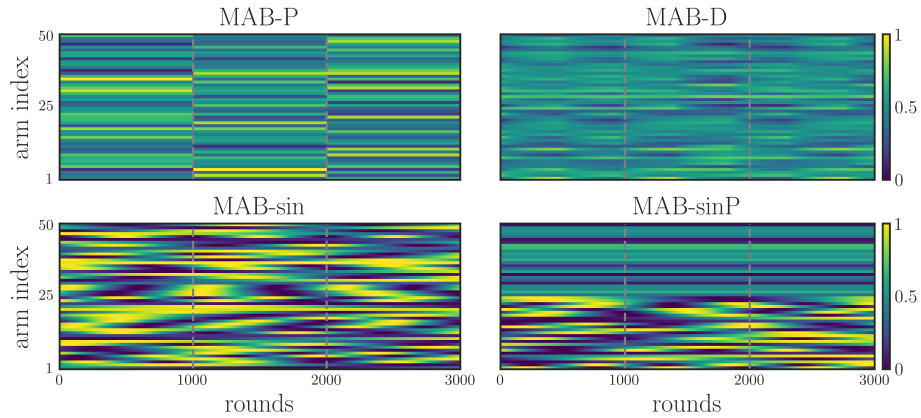


Figure 2: REWARD DISTRIBUTION FOR THE FOUR MAB VARIANTS - *The reward distribution for each variant is illustrated: piecewise stationary distribution (MAB-P), piecewise stationary distribution with drift (MAB-D), sinusoidal distribution shift (MAB-sin), partial sinusoidal distribution shift (KAB-sinP). The arms are organized in rows; the grey dashed lines demarcate trials (3), which are block of rounds represented by columns, and here only reporting the arm reward proability.*

each for $K = \{40, 200\}$, and all averaged over 2 iterations. The results are summarized below in figure 3.

sigmoid curve. This is consistent with the idea that the input of population U to population V is weighted maximally for high option values (strong synapses), whereas for weaker estimates the contributions are low or close to zero, allowing for more exploration. Interestingly, a common feature seemed to be a slight concavity after zero, a slim influence of the Gaussian component, which might be interpreted as a sort of test for newly formed synapses. However, the size of this effect is not large.

The neural response functions are shown in **3d**. Both population evolved to have a similar shape, a sharp sigmoid with a clear threshold, with population U having a more variable distribution. The form is characterized by not allowing for a fine-grained linear response but rather an high-pass filter, with activity occurring only after strong excitation. This firing behaviour is reminiscent of coincidence detector neurons, which are sometimes referred to as class III neurons with respect to their f-I curve [44].

3.3 Environment variants and number of arms

The model has been tested and compared with the other algorithms: Thompson Sampling (TS), ϵ -Greedy, and UCB. The benchmark were the four different variants of the MAB problem listed above 3.1, with a variable number of arms ranging from 5 to 1000. The results are reported in table 1. Overall, our model displays a solid performance over all environments, most of the time being equally good or better than the other algorithms. Interestingly, a large numbers of arms (K) did not present a significant challenge, as the model effectively adapted to the different environments and reward distributions. However, this is in part due to the randomness in the assignment of arm probabilities, and the statistics of the quantity of high-reward arms as their number increases. Nonetheless, given the non-stationarity of the reward distribution it is still a non trivial task to re-calibrate to new distributions.

3.4 Analysis of dynamics and robustness

3.4.1 Entropy analysis

For a better understanding of the qualitative differences between the models, we analyzed the progress over the rounds by tracking the selected arms in a simple piecewise stationary distribution environment. The simulation was run for 3 trials with 2000 rounds each, and then averaged over 5 iterations. Additionally, in order to quantify the variability of the decision policy at a given time and highlight the particularity of each decision-making behaviour, we calculated the entropy of the probability distribution p of chosen arms, calculated over a window of 20 rounds, as $H = -\sum_i^K p_i \log(p_i)$. The unit of entropy is in nats, and it ranges from 0 (no uncertainty) to $\log_e(K)$ (maximum uncertainty). In figure 4-a, it is plotted for each model the raster plot of selected arms together with its level of entropy. The reward probability distribution over the arms has an average of $H = 2.02$.

	K	5	10	50	100	200	1000
MAB-P	TS	0.03(8)	0.02(6)	0.02(7)	0.04(5)	0.02(3)	0.16(2)
	ϵ -Greedy	0.05(14)	0.07(5)	0.08(7)	0.15(6)	0.08(2)	0.10(4)
	UCB	0.05(15)	0.05(8)	0.19(6)	0.33(3)	0.39(2)	0.54(3)
	Model	<i>0.08(13)</i>	<i>0.07(11)</i>	<i>0.07(14)</i>	<i>0.07(8)</i>	<i>0.09(9)</i>	0.07(8)
MAB-D	TS	0.03(6)	0.08(13)	0.16(6)	0.19(3)	0.28(7)	0.34(3)
	ϵ -Greedy	0.04(7)	0.14(13)	0.22(5)	0.19(8)	0.26(7)	0.16(4)
	UCB	0.05(6)	0.09(13)	0.21(3)	0.36(4)	0.40(3)	0.49(2)
	Model	<i>0.13(10)</i>	<i>0.15(16)</i>	0.05(6)	<i>0.21(5)</i>	0.26(7)	0.12(7)
MAB-sin	TS	0.21(22)	0.22(16)	0.07(5)	0.10(5)	0.06(4)	0.21(5)
	ϵ -Greedy	0.21(21)	0.18(10)	0.12(5)	0.12(6)	0.10(4)	0.10(1)
	UCB	0.03(4)	0.05(3)	0.17(4)	0.23(1)	0.33(3)	0.49(3)
	Model	0.00(3)	0.02(4)	0.05(3)	0.06(4)	<i>0.08(1)</i>	0.05(4)
MAB-sinP	TS	0.19(21)	0.43(19)	0.17(10)	0.11(6)	0.09(6)	0.19(6)
	ϵ -Greedy	0.24(26)	0.43(10)	0.24(10)	0.14(5)	0.15(6)	0.14(2)
	UCB	0.00(6)	0.26(17)	0.18(6)	0.29(3)	0.34(1)	0.52(3)
	Model	0.00(9)	0.23(17)	0.14(7)	0.08(8)	0.06(4)	0.09(7)

Table 1: TABLE OF PERFORMANCE — *From top to bottom: results for MAB-P, MAB-D, MAB-sin, and MAB-sinP, for different numbers K of arms. Each cell shows average regret and standard deviation (in parentheses), computed over 2 trials of 2000 rounds, averaged over 5 simulations.*

As expected, the shape of the entropy curve expresses the inherent strategy adopted by each model. In particular, the UCB algorithm showed the highest variability, marked by a persistent exploratory behaviour throughout the trials despite converging to reward options. Thompson Sampling was able to reach most solutions, although with difficulty in adapting to new reward distributions leading to high entropy levels. ϵ -Greedy also showed a good performance quite reliably, with the greedy strategy assuring low entropy for most of the rounds. Similar behaviour was observed for our model, which was able to reach the optimal policy and maintain it over time, with entropy peaking mostly at the beginning of the trials and being, on average, the lowest among all models. Indeed, the dynamics of our model make it particularly suited for the task of non-stationary MAB, as it is able to quickly adapt to new reward distributions and firmly maintain a greedy policy.

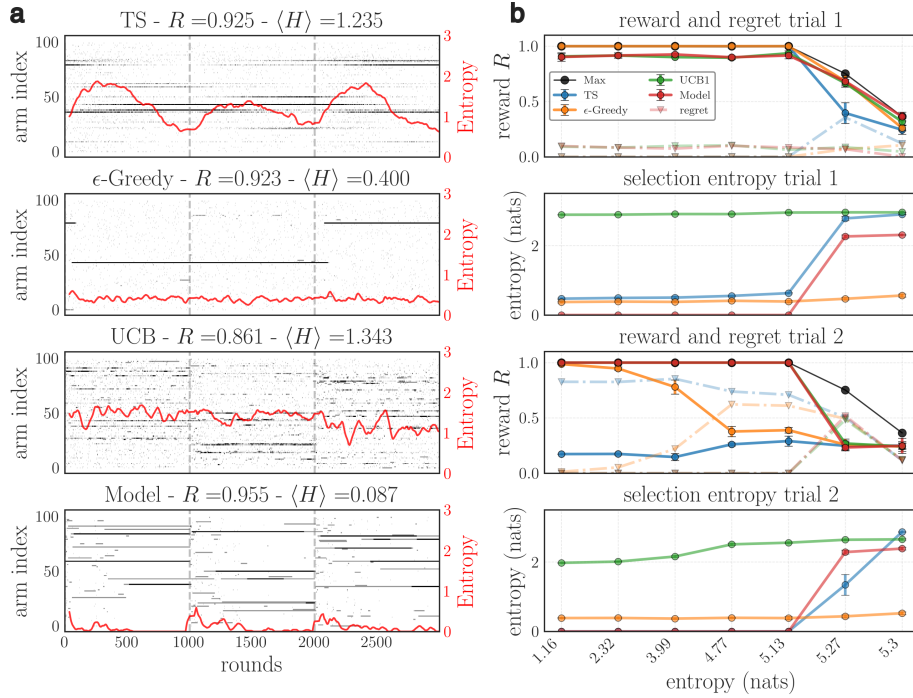


Figure 4: DECISION-MAKING DYNAMICS FOR DIFFERENT MODELS - **a**: Each plot display the results from one model. The raster plots (black dots) show the arms selected at each round. The red lines represent the entropy level, calculated from the distribution of selections over the preceeding 20 rounds, smoothed with a 30-steps moving average. In the plot titles, the total reward and average entropy (H) over all trials are also reported. - **b**: the top two rows display the average reward for trial 1 and 2 obtained by each model for increasing levels of entropy (in nats) in the reward distribution; a dashed line is the regret with respect to the upper bound (black solid line). The two bottom average entropy of the selections for the first and second trial of the simulation, each with 2000 rounds.

Then, we sought to investigate the robustness of the model by targeting the capacity to endure increasing levels of entropy in the reward distribution, defined as $-\sum_i^N p_i \log(p_i)$ and calculated in nats. In particular, this analysis tracked the performance, cumulative reward and regret, as well as the selection entropy, defined as the entropy of the distribution of selected options within a sliding window of 20 rounds.

The simulation was done in a piecewise stationary environment with $K = 200$ in two trials averaged over 5 independent runs, and it is showed in figure 4b. The distributions were chosen such to have only one strongly rewarding arm, in order to highlight the models ability to find it. For more details about the distribution see the appendix 5.4. In the top two plots, it is shown the average reward and regret obtained by each model against the reward distribution

entropy for the two trials.

The results reported how all models are capable of robust performance in the first trial even in the presence of high uncertainty. In the second trial, ϵ -Greedy and Thompson Sampling suffered the increasing difficulty of switching arms, probably due to their conservative approaches. However, this challenge afflicted UCB and our model only with higher entropy levels, recognizing their adaptability.

Another perspective to this analysis was given by the two bottom plots, which showed the average entropy over the trials. Overall, there was the unsurprising trend of increasing selection entropy with the entropy of the reward distribution. Nonetheless, striking is the exception of Epsilon-Greedy, which still maintained a constant level throughout. UCB displayed the highest average values, while Thompson Sampling followed with some delay. On the other hand, our model display a more abrupt change, going from a state of very low to high variability, sign of a solid exploratory behaviour.

4 Discussion

The ability to make decisions under uncertainty is a fundamental aspect of cognition. One well-established framework for studying this capacity is the multi-armed bandit (MAB) problem, which has been widely explored and extended across multiple domains [1, 45].

In behavioral experiments, humans demonstrate remarkable adaptability in such settings, integrating environmental uncertainty, generalizing across trials, and dynamically adjusting their learning rates. These behaviors reflect a diversity of cognitive strategies [46]. While Bayesian approaches often capture human behavior well [16, 47, 48], they are challenging to map directly onto biologically realistic neural dynamics. Despite the existence of many algorithms with strong theoretical guarantees, most lack biological plausibility—particularly in their architectural assumptions and mechanisms of learning and choice.

In this work, we aimed to design a minimal, biologically inspired architecture able of solving non-stationary MAB tasks. Specifically, we proposed a simple model composed of two interacting populations of rate-based neurons connected via plastic synapses. We evaluated the model across four variants of the MAB problem, each differing in how reward probabilities evolved over time, and across a wide range of arm counts—from a few options to over a thousand. For comparison, we also tested three standard algorithms.

Our results show demonstrated the model’s ability of adapting to changing reward distributions and quickly recovering performance over time. It reliably tracked reward-optimal options and sustained effective decision policies, matching the performance of established methods such as Thompson Sampling, ϵ -Greedy, and Upper Confidence Bound.

To better understand the behavior of the system, we analyzed its responses across varying levels of reward distribution entropy. In low-uncertainty settings, the model quickly identified the rewarding option and adopted a greedy

policy—similarly to Thompson Sampling. In contrast, UCB maintained a higher degree of exploration. As uncertainty increased, the model exhibited higher option entropy in its decisions, transitioning to a more exploratory strategy akin to UCB. While this shift modestly impaired the model’s ability to switch arms in highly volatile environments, overall performance remained robust.

The strengths of the model can be traced both in the architecture and in the learning paradigm, whose hyperparameters were optimized through an evolutionary process. Interestingly, the values found converged to solutions that can be mapped to plausible synaptic mechanisms. On one hand the neural dynamics, which rely on plastic connections and a consensus-like selection process. Particularly important was the choice of modulating the afferent connections to the value population V according to a non-linear function dependant on the synaptic weight itself. In so doing, it was possible to evolve implicitly an effective option-value policy for the tradeoff between exploration and exploitation.

The emerged neural response functions were characterized by a steep sigmoidal shape, which can be related to the saturation of the neural response once a certain threshold is crossed, a feature observed in biological network as class III neurons, besides being a common choice for artificial ones [44, 33, 34].

On another hand, learning was structured as a non-associative plasticity rule based on the reward. Similarly to before, a non-linear function of the synaptic weights played a critical role, specifically in defining the synapse-specific learning rate [17]. Further, the evolved shape of the learning rate function was inversely proportional to the synaptic weight, which can be related to the resource availability at the synapse and its state, including the size [19, 20].

An additional consideration is the inspiration from the functional role of the orbitofrontal cortex (OFC) and anterior cingulate cortex (ACC), two important pre-frontal regions known to be involved in decision-making processes [49, 50].

In our work, we have explored ways in which an option value can be formed according to recent reward history and connection weights, updating the option representations. The OFC is known to represent different options, updating their values based on reward history and outcomes [51, 49, 52].

Next, the model relies on some inductive biases -the shape of the Gaussian-sigmoid function, affecting the neural activity and the weight-dependant learning rate. These biases may be considered to encode implicitly a policy dictating how the two populations interact, how new information should be incorporated into the weight value update, and what option to choose next. Indeed, the ACC has been associated with action evaluation and regulating the balance between exploration and exploitation [50, 53].

Additionally, the generation of an option selection results from a temporal interaction of the activity of the two neural population, before converging to a choices. In a similar direction, it has been observed the OFC to transiently visit chosen and unchosen options before committing [54] Further, the dynamic interaction between the ACC and OFC has been linked to transient pre-stimulus activations, which bias decisions toward the most valuable option [55, 56, 57].

Despite the promising results, there are some limitations to the model. First and foremost, the great level of abstraction in the neuronal details, as we con-

sidered simple point neurons with synapses modeled with relatively elementary functions, lacking the anatomical complexity of actual dendrites. In particular, the model does not account for the presence of noise in the neural dynamics, which is a well-known feature of biological neurons [58]. Further, the functional association with the pre-frontal cortical region is only moderate, although present. On the computational side, since our interest lied in the biological plausibility and evolution of adaptive meta-learning solutions, we used as reference only a few well established and relatively simple algorithms, and did not take into account more advanced variants such as VDBE [9, 10], *f-Discounted-Sliding-Window Thompson Sampling (f-dsw TS)* [30], and variants of ϵ -Greedy [31].

Future work could involve the comparison with more sophisticated algorithms, the introduction of a larger architecture, and more realistic neural dynamics, such as spiking neurons [59].

Acknowledgements & Statements

The authors declare no competing interests.

The code is publicly available and can be found at <https://github.com/iKiru-hub/minBandit.git>.

This research was funded by the European Union’s Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement N^o 945371 and the University of Oslo.

The research presented in this paper has benefited from the Experimental Infrastructure for Exploration of Exascale Computing (eX3), which is financially supported by the Research Council of Norway under contract 270053.

Lastly, special thanks to Kosio Beshkov and Marianne Fyhn for inputs and feedback.

References

- [1] Richard S. Sutton and Andrew G. Barto. The Reinforcement Learning Problem. In *Reinforcement Learning: An Introduction*, pages 51–85. MIT Press, 1998.
- [2] Yael Niv, Daphna Joel, Isaac Meilijson, and Eytan Ruppin. Evolution of Reinforcement Learning in Uncertain Environments: A Simple Explanation for Complex Foraging Behaviors. *International Society for Adaptive Behavior*, 2002.
- [3] Bruno B. Averbeck. Theory of Choice in Bandit, Information Sampling and Foraging Tasks. *PLoS Computational Biology*, 11(3):e1004164, March 2015.

- [4] Shipra Agrawal and Navin Goyal. Analysis of Thompson Sampling for the Multi-armed Bandit Problem. In *Proceedings of the 25th Annual Conference on Learning Theory*, pages 39.1–39.26. JMLR Workshop and Conference Proceedings, June 2012.
- [5] Emilie Kaufmann, Nathaniel Korda, and Rémi Munos. Thompson Sampling: An Asymptotically Optimal Finite Time Analysis, July 2012.
- [6] Peter Auer and Nicolo Cesa-Bianchi. Finite-time Analysis of the Multi-armed Bandit Problem. *Machine Learning*, 2002.
- [7] J. C. Gittins. Bandit Processes and Dynamic Allocation Indices. *Journal of the Royal Statistical Society. Series B (Methodological)*, 41(2):148–177, 1979.
- [8] Yikun Ban, Jingrui He, and Curtiss B. Cook. Multi-facet Contextual Bandits: A Neural Network Perspective, June 2021.
- [9] Michel Tokic. Adaptive ε -Greedy Exploration in Reinforcement Learning Based on Value Differences. In Rüdiger Dillmann, Jürgen Beyerer, Uwe D. Hanebeck, and Tanja Schultz, editors, *KI 2010: Advances in Artificial Intelligence*, volume 6359, pages 203–210. Springer Berlin Heidelberg, Berlin, Heidelberg, 2010.
- [10] Michel Tokic and Günther Palm. Value-Difference Based Exploration: Adaptive Control between Epsilon-Greedy and Softmax. In Joscha Bach and Stefan Edelkamp, editors, *KI 2011: Advances in Artificial Intelligence*, volume 7006, pages 335–346. Springer Berlin Heidelberg, Berlin, Heidelberg, 2011.
- [11] Jangho Lee, Jeonghee Jo, Byounghwa Lee, Jung-Hoon Lee, and Sungroh Yoon. Brain-inspired Predictive Coding Improves the Performance of Machine Challenging Tasks. *Frontiers in Computational Neuroscience*, 16:1062678, 2022.
- [12] M. W. Spratling. A review of predictive coding algorithms. *Brain and Cognition*, 112:92–97, March 2017.
- [13] Samuel Schmidgall, Rojin Ziaei, Jascha Achterberg, Louis Kirsch, S. Pardis Hajiseyedrazi, and Jason Eshraghian. Brain-inspired learning in artificial neural networks: A review. *APL Machine Learning*, 2(2):021501, May 2024.
- [14] Demis Hassabis, Dharshan Kumaran, Christopher Summerfield, and Matthew Botvinick. Neuroscience-Inspired Artificial Intelligence. *Neuron*, 95(2):245–258, July 2017.
- [15] Ziming Liu, Eric Gan, and Max Tegmark. Seeing is Believing: Brain-Inspired Modular Training for Mechanistic Interpretability, June 2023.

- [16] Timothy E. J. Behrens, Mark W. Woolrich, Mark E. Walton, and Matthew F. S. Rushworth. Learning the value of information in an uncertain world. *Nature Neuroscience*, 10(9):1214–1221, September 2007.
- [17] Rylan S Larsen and P Jesper Sjöström. Synapse-type-specific plasticity in local circuits. *Current opinion in neurobiology*, 35:127–135, December 2015.
- [18] Arne V. Blackman, Therese Abrahamsson, Rui Ponte Costa, Txomin Lalanne, and P. Jesper Sjöström. Target-cell-specific short-term plasticity in local circuits. *Frontiers in Synaptic Neuroscience*, 5:11, December 2013.
- [19] Thomas M. Bartol, Cailey Bromer, Justin Kinney, Michael A. Chirillo, Jennifer N. Bourne, Kristen M. Harris, and Terrence J. Sejnowski. Hippocampal Spine Head Sizes Are Highly Precise, March 2015.
- [20] Pablo Ariel, Michael B. Hoppa, and Timothy A. Ryan. Intrinsic variability in Pv, RRP size, Ca(2+) channel repertoire, and presynaptic potentiation in individual synaptic boutons. *Frontiers in Synaptic Neuroscience*, 4:9, 2012.
- [21] Jeffrey B. Inglis, Vivian V. Valentin, and F. Gregory Ashby. Modulation of Dopamine for Adaptive Learning: A Neurocomputational Model. *Computational brain & behavior*, 4(1):34–52, March 2021.
- [22] Kiyohito Iigaya, Giles W Story, Zeb Kurth-Nelson, Raymond J Dolan, and Peter Dayan. The modulation of savouring by prediction error and its effects on choice. *eLife*, 5:e13747, April 2016.
- [23] Clara Kwon Starkweather, Samuel J. Gershman, and Naoshige Uchida. The Medial Prefrontal Cortex Shapes Dopamine Reward Prediction Errors under State Uncertainty. *Neuron*, 98(3):616–629.e6, May 2018.
- [24] Benedicte M. Babayan, Naoshige Uchida, and Samuel J. Gershman. Belief state representation in the dopamine system. *Nature Communications*, 9(1):1891, May 2018.
- [25] Peyman Khorsand and Alireza Soltani. Optimal structure of metaplasticity for adaptive learning. *PLOS Computational Biology*, 13(6):e1005630, June 2017.
- [26] Shiva Farashahi, Christopher H. Donahue, Peyman Khorsand, Hyojung Seo, Daeyeol Lee, and Alireza Soltani. Metaplasticity as a Neural Substrate for Adaptive Learning and Choice under Uncertainty. *Neuron*, 94(2):401–414.e6, April 2017.
- [27] Stefano Fusi, Patrick J. Drew, and L. F. Abbott. Cascade Models of Synaptically Stored Memories. *Neuron*, 45(4):599–611, February 2005.

- [28] Aurélien Garivier and Eric Moulines. On Upper-Confidence Bound Policies for Non-Stationary Bandit Problems, May 2008.
- [29] Omar Besbes, Yonatan Gur, and Assaf Zeevi. Stochastic Multi-Armed-Bandit Problem with Non-stationary Rewards. In *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014.
- [30] Emanuele Cavenaghi, Gabriele Sottocornola, Fabio Stella, and Markus Zanker. Non Stationary Multi-Armed Bandit: Empirical Evaluation of a New Concept Drift-Aware Algorithm. *Entropy*, 23(3):380, March 2021.
- [31] Han Qi, Fei Guo, and Li Zhu. Forced Exploration in Bandit Problems, December 2023.
- [32] Alex Martin. The Representation of Object Concepts in the Brain. *Annual Review of Psychology*, 58(1):25–45, January 2007.
- [33] Gabriel Koch Ocker and Michael A. Buice. Flexible neural connectivity under constraints on total connection strength, January 2020.
- [34] Andrea Apicella, Francesco Donnarumma, Francesco Isgrò, and Roberto Prevede. A survey on modern trainable activation functions. *Neural Networks*, 138:14–32, June 2021.
- [35] Paul Miller and Jonathan Cannon. Combined mechanisms of neural firing rate homeostasis. *Biological Cybernetics*, 113(1):47–59, 2019.
- [36] Lars Bäckman, Lars Nyberg, Anna Soveri, Jarkko Johansson, Micael Andersson, Erika Dahlin, Anna S. Neely, Jere Virta, Matti Laine, and Juha O. Rinne. Effects of Working-Memory Training on Striatal Dopamine Release. *Science*, 333(6043):718–718, August 2011.
- [37] Pierre Enel, Joni D Wallis, and Erin L Rich. Stable and dynamic representations of value in the prefrontal cortex. *eLife*, 9:e54313, July 2020.
- [38] Ami Citri and Robert C. Malenka. Synaptic Plasticity: Multiple Forms, Functions, and Mechanisms. *Neuropsychopharmacology*, 33(1):18–41, January 2008.
- [39] Mary B. Kennedy. Synaptic Signaling in Learning and Memory. *Cold Spring Harbor Perspectives in Biology*, 8(2):a016824, February 2016.
- [40] Mohammad Samavat, Thomas M. Bartol, Kristen M. Harris, and Terrence J. Sejnowski. Synaptic Information Storage Capacity Measured With Information Theory. *Neural Computation*, 36(5):781–802, April 2024.
- [41] Christian Igel, Nikolaus Hansen, and Stefan Roth. Covariance Matrix Adaptation for Multi-objective Optimization. *Evolutionary Computation*, 15(1):1–28, March 2007.
- [42] Erkki Oja. Oja learning rule. *Scholarpedia*, 3(3):3612, March 2008.

- [43] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization, January 2017.
- [44] Stéphanie Ratté, Sungho Hong, Erik De Schutter, and Steven A. Prescott. Impact of Neuronal Properties on Network Coding: Roles of Spike Initiation Dynamics and Robust Synchrony Transfer. *Neuron*, 78(5):758–772, June 2013.
- [45] Junyang Liu. Comprehensive Exploration and Implementation of Multi-Armed Bandit Algorithms Across Various Domains. *Highlights in Science, Engineering and Technology*, 94:230–235, April 2024.
- [46] Mark Steyvers, Michael D. Lee, and Eric-Jan Wagenmakers. A Bayesian analysis of human decision-making on bandit problems. *Journal of Mathematical Psychology*, 53(3):168–179, June 2009.
- [47] Eric Schulz, Nicholas T. Franklin, and Samuel J. Gershman. Finding structure in multi-armed bandits. *Cognitive Psychology*, 119:101261, June 2020.
- [48] Shunan Zhang and Angela J Yu. Forgetful Bayes and myopic planning: Human learning and decision-making in a bandit setting. In *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc., 2013.
- [49] Steven W. Kennerley and Mark E. Walton. Decision Making and Reward in Frontal Cortex. *Behavioral Neuroscience*, 125(3):297–317, June 2011.
- [50] Mehdi Khamassi, Pierre Enel, Peter Ford Dominey, and Emmanuel Procyk. Chapter 22 - Medial prefrontal cortex and the adaptive regulation of reinforcement learning parameters. In V. S. Chandrasekhar Pammi and Narayanan Srinivasan, editors, *Progress in Brain Research*, volume 202 of *Decision Making*, pages 441–464. Elsevier, January 2013.
- [51] Chung-Hay Luk and Jonathan D. Wallis. Choice Coding in Frontal Cortex during Stimulus-Guided or Action-Guided Decision-Making. *Journal of Neuroscience*, 33(5):1864–1871, January 2013.
- [52] Miriam Cornelia Klein-Flügge, Helen Catharine Barron, Kay Henning Brodersen, Raymond J. Dolan, and Timothy Edward John Behrens. Segregated Encoding of Reward–Identity and Stimulus–Reward Associations in Human Orbitofrontal Cortex. *Journal of Neuroscience*, 33(7):3202–3211, February 2013.
- [53] Nils Kolling, Marco K. Wittmann, Tim E. J. Behrens, Erie D. Boorman, Rogier B. Mars, and Matthew F. S. Rushworth. Value, search, persistence and model updating in anterior cingulate cortex. *Nature Neuroscience*, 19(10):1280–1285, October 2016.
- [54] Erin L. Rich and Jonathan D. Wallis. Decoding subjective decisions from orbitofrontal cortex. *Nature Neuroscience*, 19(7):973–980, July 2016.

- [55] Shintaro Funahashi. Prefrontal Contribution to Decision-Making under Free-Choice Conditions. *Frontiers in Neuroscience*, 11, July 2017.
- [56] Encarni Marcos and Aldo Genovesio. Determining Monkey Free Choice Long before the Choice Is Made: The Principal Role of Prefrontal Neurons Involved in Both Decision and Motor Processes. *Frontiers in Neural Circuits*, 10, September 2016.
- [57] Zuzanna Z. Balewski, Thomas W. Elston, Eric B. Knudsen, and Joni D. Wallis. Value dynamics affect choice preparation during decision-making. *Nature neuroscience*, 26(9):1575–1583, September 2023.
- [58] A. Aldo Faisal. Noise in Neurons and Other Constraints. In N. Le Novère, editor, *Computational Systems Neurobiology*, pages 227–257. Springer Netherlands, Dordrecht, 2012.
- [59] João D. Nunes, Marcelo Carvalho, Diogo Carneiro, and Jaime S. Cardoso. Spiking Neural Networks: A Survey. *IEEE Access*, 10:60738–60764, 2022.

5 Appendix

5.1 Neural response function

The activation functions applied to the two neuronal population are defined as a step-function composed with a generalized sigmoid as follows:

$$f(x; g, o, \theta) = \begin{cases} [1 + e^{-g(x-o)}]^{-1} & \text{if } [1 + e^{-g(x-o)}]^{-1} > \theta \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Where:

- x is the neuron pre-activation value
- g is the gain
- o is the offset
- θ is the threshold

Each population has its own set of parameters, which are optimized through evolutionary search.

5.2 Gaussian-sigmoid function

The function Φ is defined by combining a generalized version of the sigmoid, namely with a gain $\beta \neq 1$ and offset $\alpha \neq 0$, and a Gaussian with mean μ and variance σ^2 . Their contributions are weighted by r and $1 - r$ ($r \in (0, 1)$) respectively.

$$\Phi_v(x) = r \left(1 + \exp^{-\beta(x-\alpha)} \right)^{-1} + (1 - r) \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$$

The motivation behind this choice is to express a function that possesses a bounded region (depending on μ, σ) at a high/low peak (depending on the value of γ_2), and a continuous transition to a constant value (depending on the steepness of the sigmoid β , shift α , and intensity γ_1).

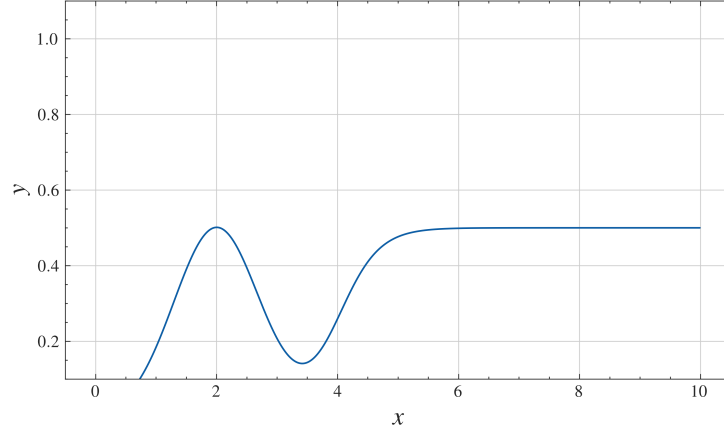


Figure 5: ACTIVATION FUNCTION Φ_v - Parameters $\beta = 10$, $\alpha = 1$, $\mu = 1$, $\sigma = 1$, and $r = 0.5$.

5.3 Evolution search

The optimization was carried out over several parameters concerning the model architecture and dynamics:

Network parameters

- τ_u : time constant of population U
- τ_v : time constant of population V
- g_u : gain of the neural response function of population U
- g_v : gain of the neural response function of population V
- o_u : offset of the neural response function of population U
- o_v : offset of the neural response function of population V
- θ_u : threshold of the neural response function of population U
- θ_v : threshold of the neural response function of population V
- W^+ : maximal weight value for the weights $\mathbf{W}^{UV}$

Option value function parameters

- β_v : steepness of the sigmoid
- α_v : shift of the sigmoid
- μ_v : mean of the Gaussian
- σ_v : variance of the Gaussian

- r_v : weight of the sigmoid

Learning rate function parameters

- β_η : steepness of the sigmoid
- α_η : shift of the sigmoid
- μ_η : mean of the Gaussian
- σ_η : variance of the Gaussian
- r_η : weight of the sigmoid

Each individual has been evaluated over environment the following environments:

- MAB-0: average reward distribution entropy $\langle H \rangle = 2.05$
- KAB-sinP: average reward distribution entropy $\langle H \rangle = 2.1$, given K arm frequencies f_k as an equally spaced set $\{0.1 \dots i \dots 0.4\}$, phases λ_k drawn from an uniform $\sim \mathcal{U}(0, 2\pi)$, and half of the arms have been set to constant values drawn from another uniform $\sim \mathcal{U}(0.1, 0.7)$; the final reward distribution was not normalized.

The number of arms was $K = 10$ and 150 , and lasted for 2 trials with 2000 rounds each. The final fitness was the average over 2 iterations.

The optimization has been implemented in Python using the **DEAP** library, and the algorithm used was the **CMA-ES** algorithm. The optimization involved 40 generations with a population size of 256 individuals. The mutation rate was set to 0.5 with a sigma of 0.8, the cross-over rate was set to 0.4. The run were carried out on a 256-core AMD EPYC 7763 with 2TB of RAM.

5.3.1 Genome distribution

Following the evolution search, it is taken the distribution of parameters over the top-scoring half of the population, corresponding to 128 individuals.

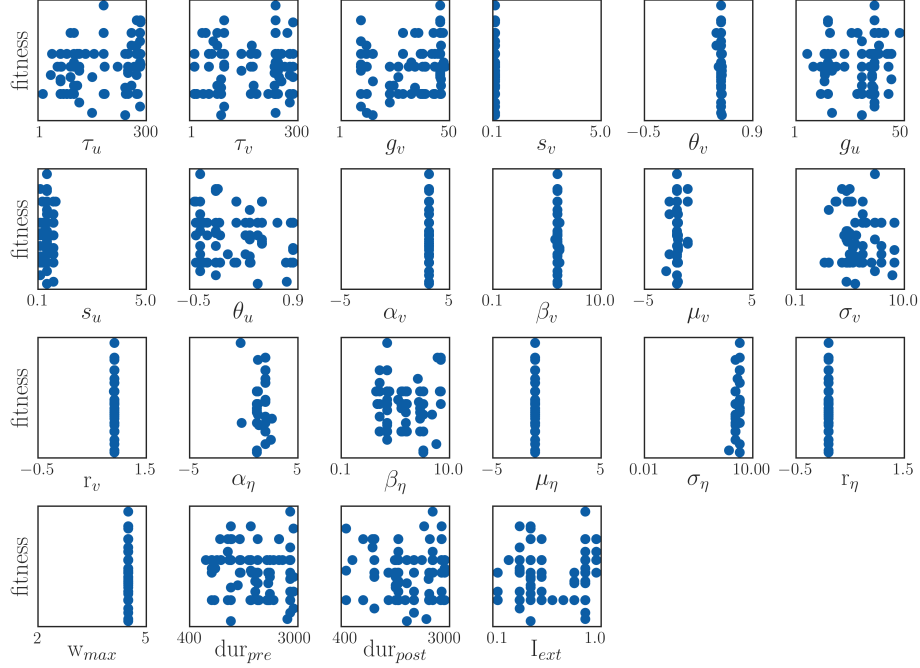


Figure 6: GENOME DISTRIBUTION - *each parameter is plotted against the fitness score.*

Figure 6 reveals those parameters shared among the models with the highest fitness score, and those that are more variable. The most stable parameters are, predictably, those involved in directly shaping the neural activity (neural activation functions) and learning policy (Gaussian sigmoid).

5.4 Reward distribution entropy

The calculation of a set of N reward probability distribution $\mathbf{p}_i$ for $i \dots N$ for K values with a progressively decreasing levels of entropy $\mathbf{h}_i$ for $i \dots N$ has been obtained by the algorithm below 5.4.

Algorithm 2: Reward Probability Distribution Generation

Input: Number of distributions N , dimension K
Output: Set of probability distributions $\mathbf{p}_i$ with decreasing entropy
Initial Setup: Define set $B = \{17, 15, 12, 8, 4, 1.5, 0.5\}$;
for $i \leftarrow 1$ **to** N **do**
 $\mathbf{z} \leftarrow \text{RandomVector} \sim \mathcal{U}(0, 0.5)^K$;
 $j \leftarrow \text{RandomIndex}(K)$;
 $\mathbf{z}_j \leftarrow 1$;
 $\beta_i \leftarrow \text{Sample index}=i \text{ from } (B)$; // Sample temperature from B
 $\mathbf{p}_i \leftarrow \frac{\exp(\beta_i \mathbf{z})}{\sum_j \exp(\beta_i \mathbf{z}_j)}$; // Softmax with temperature
end
return $\mathbf{p}_i$

5.5 Weight update dynamics

We also analyzed the weight update dynamics of the model over the rounds. In figure 7, we plotted the evolution of the total weight ΔW^{UV} over time, averaged over 20 simulations and smoothed over 30 rounds. The results show that the model is able to quickly adapt to new reward distributions. It is also able to maintain the optimal policy over time, with the weights remaining approximately stable. The update quantity ΔW_k^{UV} , which at each round is applied to one connection k , changes sign according to the collected reward, with its magnitude being higher at the beginning of the trials. Initially, the sign is mostly positive (potentiation) since the weights start at zero, and after some uncertainty a consistently preferred arm emerges. However, when the reward distribution switches a regular series of sub-optimal choices with respect to the new distribution is made, leading to zero reward. This causes an accumulation of weight updates with negative sign (depression), eventually bringing the value of the preferred arm to drop. In the meantime, other options are probed until another sequence of choices converges to another arm, promoted by a trail of positive weight updates.

This behaviour is consistent with the low entropy levels observed in the previous analysis.

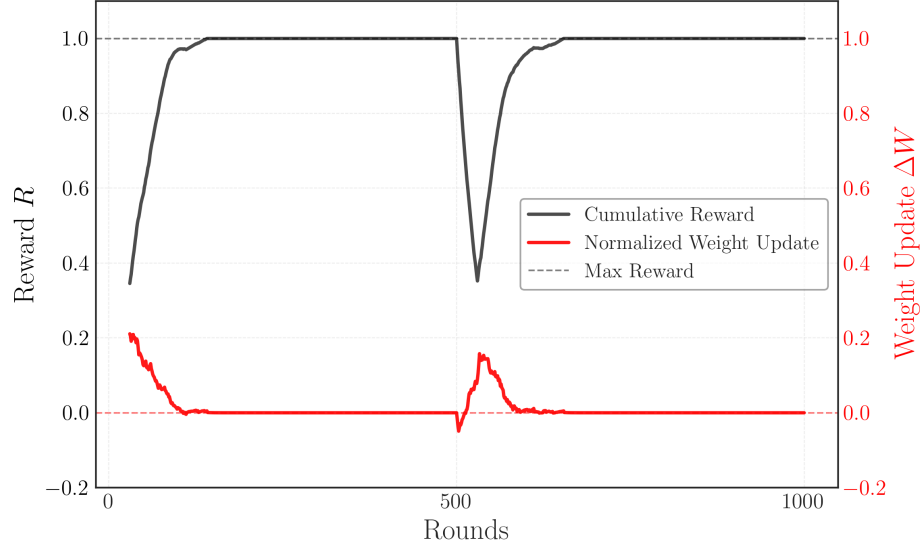


Figure 7: WEIGHT UPDATE DEVELOPEMENT FOR THE MODEL *The plot displays the weight update quantity ΔW_k^{UV} for each round (blue line), smoothed as a 20-steps moving average. It is also reported the average reward in a window of 30 rounds (orange line). The results have been obtained averaging over 30 iterations.*